Lab 3 session

Local Binary Patterns:

Target of the session:

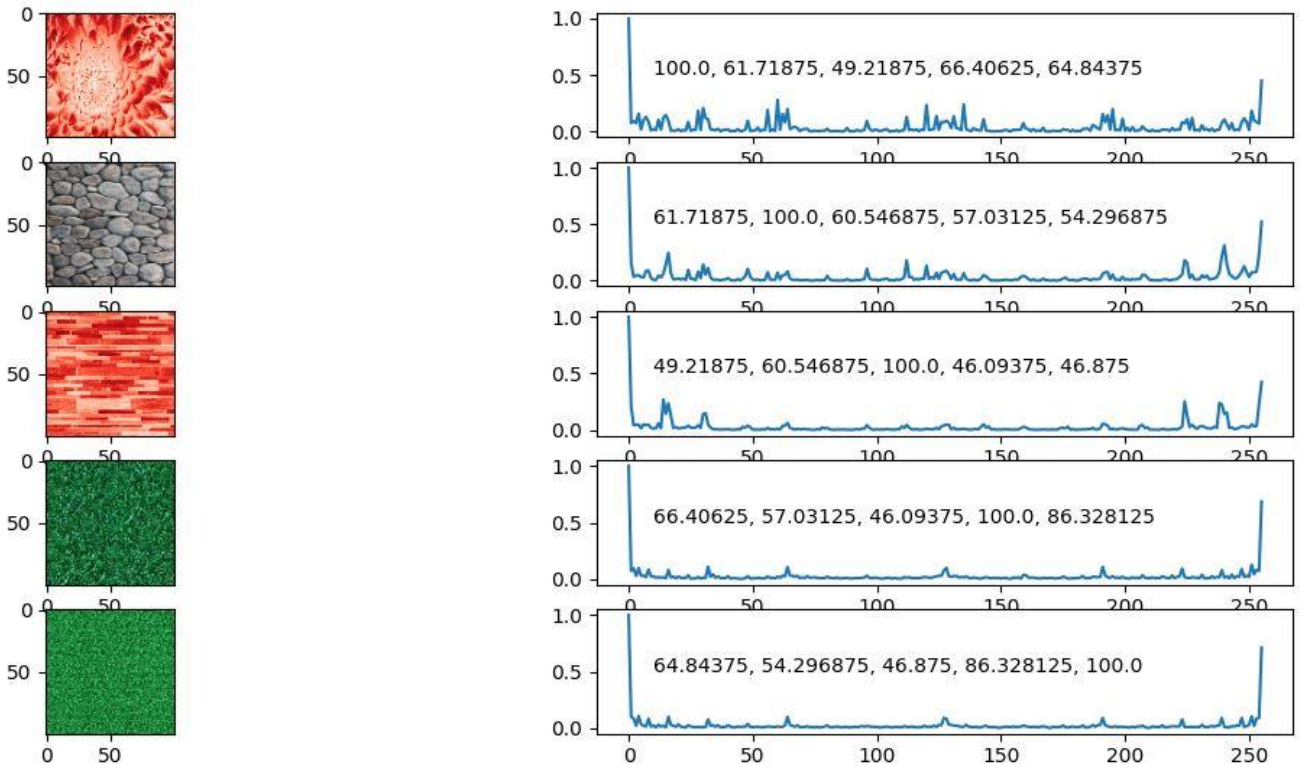


Fig. 1 final results of the session

Algorithm steps:

1. Read all of the input images in the gray scale.
2. Resize the images dimensions to become like [100, 100]pixels.
3. Extract LBP features from all input images.
4. Normalize LBP features vectors in the range [0, 1]
5. Calculate the similarities between the features vectors using Hamming distance.
6. Display your results as in the figure (1).

Functions needing to manually build from scratch:

1. Converting from binary into decimal value function.
2. LBP features extraction function.
3. Calculation of histogram of a gray image function.
4. Calculation of Hamming distance function.

Useful Explanations:

Classical LBP method:

100	50	200
191	190	220
150	40	250



0	0	1
1		1
0	0	1

N1	N2	N3
N8		N4
N7	N6	N5



Code = $(00111001)_{bin} = (57)_{dec}$

Hamming distance (using a threshold = 25):

Vector 1	Vector 2	Decimal Diff	Binary diff
150	60	90	0
120	100	20	1
60	80	20	1

So, similarity between V1 and V2 = 0.667.

Extra Task(s):
Apply the Rotation invariant LBP algorithm on some rotated images of one of the classes, and ensure that the algorithm can increase the similarity between the rotated input image and the pre-defined class.

Prepared by : Engr. Ali Ali
2023/2024
2nd term – Robotic and I.S. engineering.

----------End of session----------